

Inngest RAG App Demo

A production-ready **Retrieval-Augmented Generation (RAG)** application using **Inngest** for serverless event-driven workflows, **Qdrant** for vector storage, and **Streamlit** for the UI.

· Features

- **PDF Ingestion** · Load, chunk, embed, and store PDF documents
 - **Vector Search** · Qdrant-powered semantic search over documents
 - **LLM Generation** · OpenAI answers questions with retrieved context
 - **Throttling** · Rate-limited workflows via Inngest
 - **Streamlit UI** · Web interface for queries
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.. Architecture

```
PDF Upload · Inngest Event · [Load PDF · Chunk · Embed · Upsert to Qdrant]
                               (throttled, rate-limited)
User Query · Inngest Event · [Vector Search · Retrieve Context · OpenAI LLM
                               · Answer]
```

· File Structure

```
| File | Purpose | |-----|-----| | main.py | FastAPI + Inngest functions (ingest + query) | | data_loader.py |
PDF loading and text chunking | | vector_db.py | Qdrant vector store integration | | custom_types.py |
Pydantic models (RAGChunksAndSrc, RAGQueryResult, etc.) | | streamlit_app.py | Streamlit web UI |
```

· Getting Started

```
cd Inngest-RAG-App-Demo
uv sync
# Start API server
uv run python main.py
# Start Streamlit UI (separate terminal)
uv run streamlit run streamlit_app.py
```

Environment Variables (.env)

```
OPENAI_API_KEY=sk-...
QDRANT_URL=...
QDRANT_API_KEY=...
```

· Logic Flow

Ingestion Pipeline

- 1 **PDF loaded** · Text extracted page-by-page
- 2 **Chunking** · Text split into manageable chunks
- 3 **Embedding** · Chunks embedded via OpenAI embeddings
- 4 **Upsert** · Vectors stored in Qdrant collection

Query Pipeline

- 1 **User query** · Embedded into vector
- 2 **Search** · Top-k similar chunks retrieved from Qdrant
- 3 **Context** · Retrieved chunks assembled as context
- 4 **LLM** · OpenAI generates answer using context
- 5 **Response** · Answer displayed in Streamlit

Inngest Configuration

- **Throttle:** 2 requests per minute
 - **Rate Limit:** 1 per 4 hours (for ingestion)
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· Dependencies

fastapi, inngest, langchain, qdrant-client, openai, python-dotenv

· License

Educational project · use freely for learning and reference.